In [ ]:
```
 Experiment 4: Data Manipulation with Pandas
PIN:241UDS0097

Aim:-
Practice data manipulation using Pandas: import a CSV, set an index, inspect data, select rows and columns, create calculated columns, filter records, sort values, update data, and rename columns.
```

In [ ]:
```
## Step 1 – Import Pandas Library
```

In [1]:
```
import pandas as pd
```

In [ ]:
```
## Step 2 – Import the Dataset
```

In [2]:
```
cafe_df_241UDS0097 = pd.read_csv(
    "Experiment_4_241UDS0097_Cafe_Orders.csv"
)
cafe_df_241UDS0097.head(5)
```

Out[2]:

Columns 1–8 of 11

| | Order_ID | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent |
|---|---|---|---|---|---|---|---|
| **0** | C2001 | Cappuccino | Beverage | Vijayawada | 34 | 160 | 5.0 |
| **1** | C2002 | Veg Sandwich | Snacks | Guntur | 58 | 120 | 8.0 |
| **2** | C2003 | Iced Tea | Beverage | Visakhapatnam | 41 | 90 | 0.0 |
| **3** | C2004 | Chocolate Muffin | Bakery | Hyderabad | 27 | 110 | 10.0 |
| **4** | C2005 | Pasta Bowl | Meals | Vijayawada | 22 | 240 | 7.5 |

Columns 9–11 of 11 · first column repeated

| | Customer_Rating | Stock_Available | Payment_Method |
|---|---|---|---|
| **0** | 4.6 | 18 | UPI |
| **1** | 4.3 | 25 | Card |
| **2** | 4.1 | 32 | Cash |
| **3** | 4.5 | 14 | UPI |
| **4** | 4.4 | 11 | Card |

In [ ]:
```
## Step 3 – Set Index for the DataFrame
```

In [3]:
```
cafe_df_241UDS0097 = cafe_df_241UDS0097.set_index("Order_ID")
cafe_df_241UDS0097.head()
```

Out[3]: Columns 1–8 of 10

| | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **Order_ID** | | | | | | | |
| **C2001** | Cappuccino | Beverage | Vijayawada | 34 | 160 | 5.0 | 4.6 |
| **C2002** | Veg Sandwich | Snacks | Guntur | 58 | 120 | 8.0 | 4.3 |
| **C2003** | Iced Tea | Beverage | Visakhapatnam | 41 | 90 | 0.0 | 4.1 |
| **C2004** | Chocolate Muffin | Bakery | Hyderabad | 27 | 110 | 10.0 | 4.5 |
| **C2005** | Pasta Bowl | Meals | Vijayawada | 22 | 240 | 7.5 | 4.4 |

Columns 9–10 of 10 · first column repeated

| | Stock_Available | Payment_Method |
|---|---|---|
| **Order_ID** | | |
| **C2001** | 18 | UPI |
| **C2002** | 25 | Card |
| **C2003** | 32 | Cash |
| **C2004** | 14 | UPI |
| **C2005** | 11 | Card |

In [ ]:
```
## Step 4 – Info of the DataFrame
```

In [4]:
```
cafe_df_241UDS0097.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 20 entries, C2001 to C2020
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   Item_Name         20 non-null     object 
 1   Category          20 non-null     object 
 2   City              20 non-null     object 
 3   Units_Sold        20 non-null     int64  
 4   Unit_Price        20 non-null     int64  
 5   Discount_Percent  20 non-null     float64
 6   Customer_Rating   20 non-null     float64
 7   Stock_Available   20 non-null     int64  
 8   Payment_Method    20 non-null     object 
dtypes: float64(2), int64(3), object(4)
memory usage: 1.6+ KB
```

In [ ]:
```
## Step 5 – Columns & Rows Selection: Select a Single Column
```

In [5]:
```
cafe_df_241UDS0097["Item_Name"]
```

Out[5]:
```
Order_ID
C2001          Cappuccino
C2002        Veg Sandwich
C2003            Iced Tea
C2004    Chocolate Muffin
C2005          Pasta Bowl
C2006         Masala Chai
C2007        Cheese Toast
C2008      Blueberry Tart
C2009         Cold Coffee
C2010          Fruit Bowl
C2011             Brownie
C2012         Paneer Wrap
C2013          Lemon Soda
C2014          Veg Burger
C2015        Garlic Bread
C2016         Banana Cake
C2017       Filter Coffee
C2018        Chicken Roll
C2019         Granola Cup
C2020         Carrot Cake
Name: Item_Name, dtype: object
```

In [ ]:
```
## Step 6 – Selecting Multiple Columns
```

In [6]:
```
cafe_df_241UDS0097[["Item_Name", "Category", "City"]].head(3)
```

Out[6]:

| | Item_Name | Category | City |
|---|---|---|---|
| **Order_ID** | | | |
| **C2001** | Cappuccino | Beverage | Vijayawada |
| **C2002** | Veg Sandwich | Snacks | Guntur |
| **C2003** | Iced Tea | Beverage | Visakhapatnam |

In [ ]:
```
## Step 7 – Select Row Using iloc
```

In [7]:
```
cafe_df_241UDS0097.iloc[1]
```

Out[7]:
```
Item_Name           Veg Sandwich
Category                 Snacks
City                     Guntur
Units_Sold                   58
Unit_Price                  120
Discount_Percent            8.0
Customer_Rating             4.3
Stock_Available              25
Payment_Method             Card
Name: C2002, dtype: object
```

In [ ]:
```
## Step 8 — Select Row Using loc
```

In [8]:
```
cafe_df_241UDS0097.loc["C2020"]
```

Out[8]:
```
Item_Name           Carrot Cake
Category                 Bakery
City                  Vijayawada
Units_Sold                    21
Unit_Price                   155
Discount_Percent             7.0
Customer_Rating              4.6
Stock_Available                8
Payment_Method               UPI
Name: C2020, dtype: object
```

In [ ]:
```
## Step 9 — Inspect the First Two Rows
```

In [9]:
```
cafe_df_241UDS0097.head(2)
```

Out[9]:

Columns 1–8 of 10

| | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **Order_ID** | | | | | | | |
| **C2001** | Cappuccino | Beverage | Vijayawada | 34 | 160 | 5.0 | 4.6 |
| **C2002** | Veg Sandwich | Snacks | Guntur | 58 | 120 | 8.0 | 4.3 |

Columns 9–10 of 10 · first column repeated

| | Stock_Available | Payment_Method |
|---|---|---|
| **Order_ID** | | |
| **C2001** | 18 | UPI |
| **C2002** | 25 | Card |

In [ ]:
```
## Step 10 — Creating Total Sales Column
```

In [ ]:
```
cafe_df_241UDS0097["Total_Sales_241UDS0097"] = (
    cafe_df_241UDS0097["Units_Sold"] * cafe_df_241UDS0097["Unit_Price"]
)
cafe_df_241UDS0097.head(2)
```

In [ ]:
```
## Step 11 — Creating Discount Amount Column
```

In [ ]:
```
cafe_df_241UDS0097["Discount_Amount_241UDS0097"] = (
    cafe_df_241UDS0097["Total_Sales_241UDS0097"]
    * cafe_df_241UDS0097["Discount_Percent"] / 100
)
cafe_df_241UDS0097.head(2)
```

In [ ]:
```
## Step 12 — Creating Net Sales Column
```

In [ ]:
```
cafe_df_241UDS0097["Net_Sales_241UDS0097"] = (
    cafe_df_241UDS0097["Total_Sales_241UDS0097"]
    - cafe_df_241UDS0097["Discount_Amount_241UDS0097"]
)
cafe_df_241UDS0097.head(2)
```

In [ ]:
```
## Step 13 — Round Calculated Values
```

In [ ]:
```
cafe_df_241UDS0097["Discount_Amount_241UDS0097"] = (
    cafe_df_241UDS0097["Discount_Amount_241UDS0097"].round(2)
)
cafe_df_241UDS0097["Net_Sales_241UDS0097"] = (
    cafe_df_241UDS0097["Net_Sales_241UDS0097"].round(2)
)
cafe_df_241UDS0097.head(2)
```

In [ ]:
```
## Step 14 — Conditional Filtering: Units Sold Greater Than 40
```

In [ ]:
```
cafe_df_241UDS0097[
    cafe_df_241UDS0097["Units_Sold"] > 40
].head(5)
```

In [ ]:
```
## Step 15 — High Sales AND Low Stock
```

In [ ]:
```
cafe_df_241UDS0097[
    (cafe_df_241UDS0097["Total_Sales_241UDS0097"] > 5000)
    & (cafe_df_241UDS0097["Stock_Available"] < 20)
].head(5)
```

In [ ]:
```
## Step 16 – Products from Vijayawada OR Hyderabad
```

In [ ]:
```
cafe_df_241UDS0097[
    (cafe_df_241UDS0097["City"] == "Vijayawada")
    | (cafe_df_241UDS0097["City"] == "Hyderabad")
].head()
```

In [ ]:
```
## Step 17 – Rating at Least 4.5 AND Stock Below 25
```

In [ ]:
```
cafe_df_241UDS0097[
    (cafe_df_241UDS0097["Customer_Rating"] >= 4.5)
    & (cafe_df_241UDS0097["Stock_Available"] < 25)
]
```

In [ ]:
```
## Step 18 – Filter Multiple Cities Using `.isin()`
```

In [ ]:
```
cafe_df_241UDS0097[
    cafe_df_241UDS0097["City"].isin(["Vijayawada", "Bengaluru", "Chennai"])
].head()
```

In [ ]:
```
## Step 19 – Filter Multiple Categories Using `.isin()`
```

In [ ]:
```
cafe_df_241UDS0097[
    cafe_df_241UDS0097["Category"].isin(["Beverage", "Bakery"])
].head()
```

In [ ]:
```
## Step 20 – Filter Multiple Payment Methods Using `.isin()`
```

In [ ]:
```
cafe_df_241UDS0097[
    cafe_df_241UDS0097["Payment_Method"].isin(["UPI", "Card"])
].head()
```

In [ ]:
```
## Step 21 – Sort by Units Sold (Descending)
```

In [ ]:
```
cafe_df_241UDS0097.sort_values(
    "Units_Sold", ascending=False
).head()
```

In [ ]:
```
## Step 22 – Sort by Unit Price (Ascending)
```

In [ ]:
```
cafe_df_241UDS0097.sort_values(
    "Unit_Price", ascending=True
).head()
```

In [ ]:
```
## Step 23 – Sort by Multiple Columns
```

In [ ]:
```
cafe_df_241UDS0097.sort_values(
    ["Category", "Units_Sold"], ascending=[True, False]
).head()
```

In [ ]:
```
## Step 24 — Update Data in the Table
```

In [ ]:
```
cafe_df_241UDS0097["City"] = cafe_df_241UDS0097["City"].replace(
    {"Vijayawada": "VJA"}
)
cafe_df_241UDS0097[cafe_df_241UDS0097["City"] == "VJA"].head(2)
```

In [ ]:
```
## Step 25 — Update a Specific Item Value
```

In [ ]:
```
cafe_df_241UDS0097.loc["C2001", "Unit_Price"] = 175
print(cafe_df_241UDS0097.loc["C2001", "Unit_Price"])
```

In [ ]:
```
## Step 26 — Change Column Names
```

In [ ]:
```
cafe_df_241UDS0097 = cafe_df_241UDS0097.rename(
    columns={
        "Item_Name": "Product_Name",
        "City": "Location",
        "Customer_Rating": "Rating",
        "Net_Sales_241UDS0097": "Revenue"
    }
)
cafe_df_241UDS0097.head(1)
```

In [ ]:
```
## Step 27 — Final DataFrame
```

In [ ]:
```
cafe_df_241UDS0097.head()
```

In [ ]:
```
## Step 28 — Final Dataset Information
```

In [ ]:
```
cafe_df_241UDS0097.info()
```

In [ ]:
```
## Result
Pandas data manipulation was performed on a cafe sales dataset, including
row and column selection, calculated sales and discount columns, conditional
filtering, sorting, updating values, and renaming columns.
```